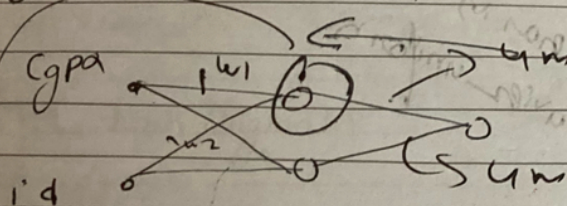


Batch Normalization (BN)

It is algorithmic method used to make the training of Deep Neural Networks faster & more stable

- 1) It uses min batch GID
- 2) layer by layer



$$Z_{11} = w_1 cgra + w_2 id + b$$

$$\begin{aligned} 1) & Z_{11} \rightarrow Z_{11}^N \rightarrow Z_{11}^{NB} \rightarrow g(Z_{11}^{NB}) = a_{11} \\ 2) & Z_{11} \rightarrow g(Z_{11}) \rightarrow g(Z_{11})^N \rightarrow g(Z_{11})^{NB} = a_{11} \end{aligned}$$

$$Z_{11}^N = Z_{11} - \mu \rightarrow \text{mean}$$

→ Standard deviation

$$\mu = \frac{1}{m} \sum_{i=1}^m Z_{11}^i$$

↑
Batch

cgra	id	Bh
8.9	100	1
6.2	89	0
7.7	18	0
9.1	91	1
⋮	⋮	⋮
⋮	⋮	⋮

$$\sigma = \sqrt{\frac{1}{m} \sum_{i=1}^m (Z_{11}^i - \mu)^2}$$

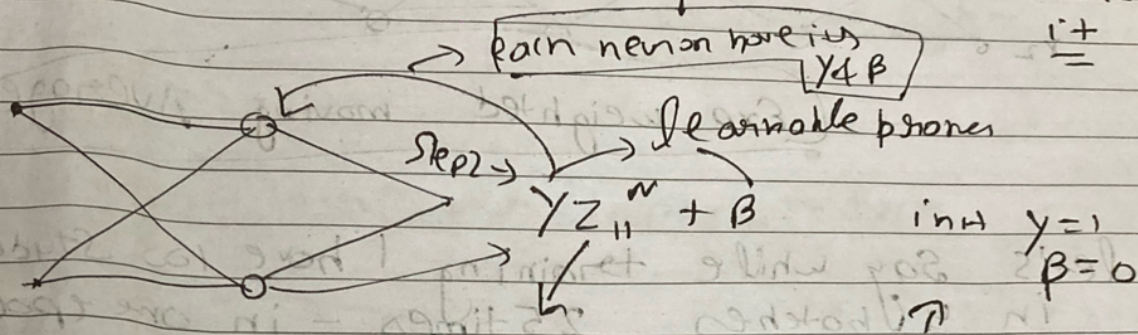
(4,2) * (2,2) → size of WW

(4,2) + (4,2) → Bias = (4,2)

for every neuron we will calculate mean for each

Step 1 $\rightarrow z_{ii}^i = \frac{z_{ii}^i - \mu_B}{\sigma_B + \epsilon}$

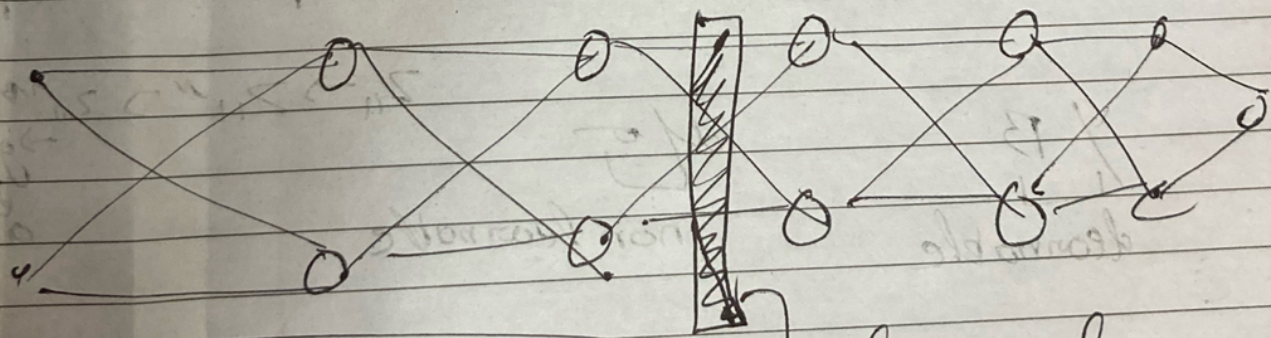
Error term. if $\epsilon = 0$ then to handle



$z_i \rightarrow z_{ii}^{nn} \rightarrow z_{ii}^{BN} \rightarrow g(z_{ii}^{BN})$ Change of the bias

perform for all neurons

$\gamma, \beta \rightarrow$ provides flexibility in NN if they need normalization or not.



$$\gamma = \gamma - \eta \frac{\partial L}{\partial \gamma} \rightarrow \gamma \text{ update}$$

unlearnable param

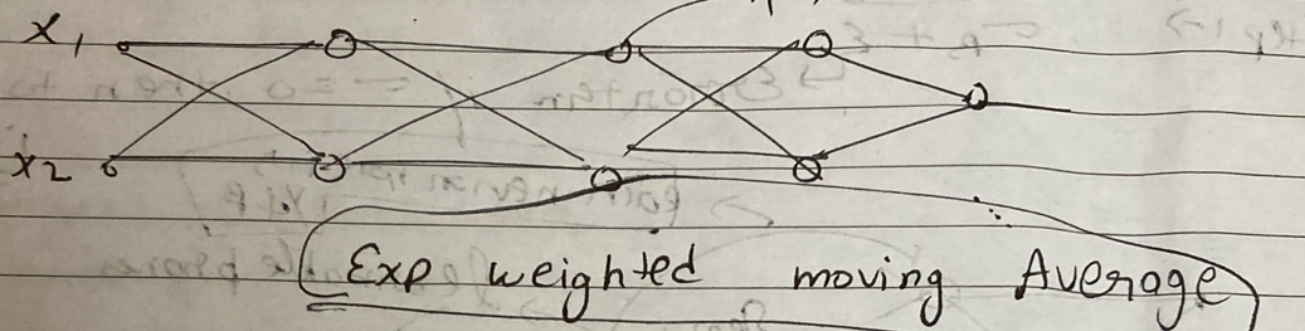
γ, β

for 1st neuron

γ, β

for 2nd neuron

Batch Norm during test



Let's say while training I have 100 students in 4 batches, 2.5 times - in one epoch

1 batch $\rightarrow \mu_{B_1}$
2 Batch $\rightarrow \mu_{B_2}$
...

μ_{25} $\rightarrow$ helps in predict and follow stores last update seq

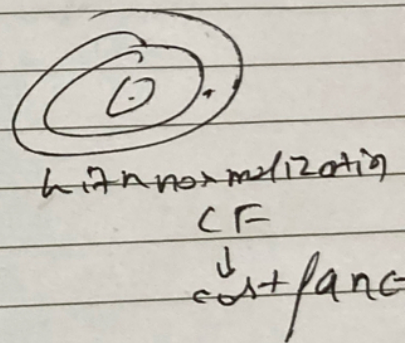
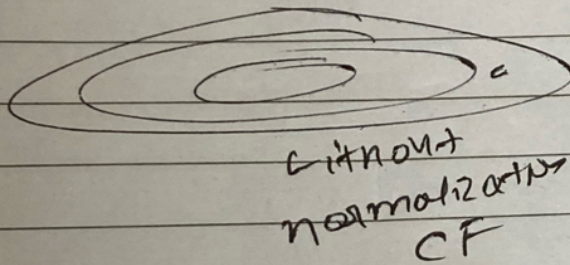
$\frac{1}{B}$
learnable

μ_{σ}
non learnable

$z_n \rightarrow z_{n+1} \rightarrow z_{n+2}$
 $\rightarrow g(z_n)$
 a_n

Advantages

- 1) Stable $\rightarrow$ hyper parameter $\rightarrow$ wider range of values
- 2) training become fast $\rightarrow$ lr can be higher
- 3) Regularization
- 4) Weight Initialization important measure



model.add(BatchNormalization())